



UNIVERSITY
OF TRENTO - Italy

Dipartimento di Ingegneria e Scienza dell'Informazione



LoE

The Logic of Entities

(HP2T)

Oct 13, 2023

LoE – The Logic of entities

- **Intuition**
- Definition
- Domain
- Language
- Interpretation function
- Theories
- Entailment
- Reasoning problems
- Key notions

LoE – Why a logic of entities?

- The **Logic of Entities (LoE)** allows us to describe entities as a function of the main class to which they belong, and their properties and relations.
- LoE formalizes and has the same level of expressiveness and reasoning capabilities as relational DBs and KGs which have entities as nodes (entity graphs).
- It automates, in a provably correct way, the kind of reasoning which is implemented in DBs and KGs. In particular, it allows for checking the facts which hold of entities when interacting among them.
- It the logic for reasoning about **entity graphs** (only)

LoE – Highlights

- All entities are assumed to have a name which uniquely identifies them, which may be unknown (multiple names for the same entity being allowed).
- It is world logic, with a graph linguistic / analogic representation.
- The LoE language has the same expressibility as EG graphs. It provides basic DB/KG reasoning capabilities. But it provides the basis for reasoning about composite world models
- It is conceptually similar in spirit to an earlier logic called the Assertion Box (ABOX) of Description Logics (DL). The move is from DBs to KGs.

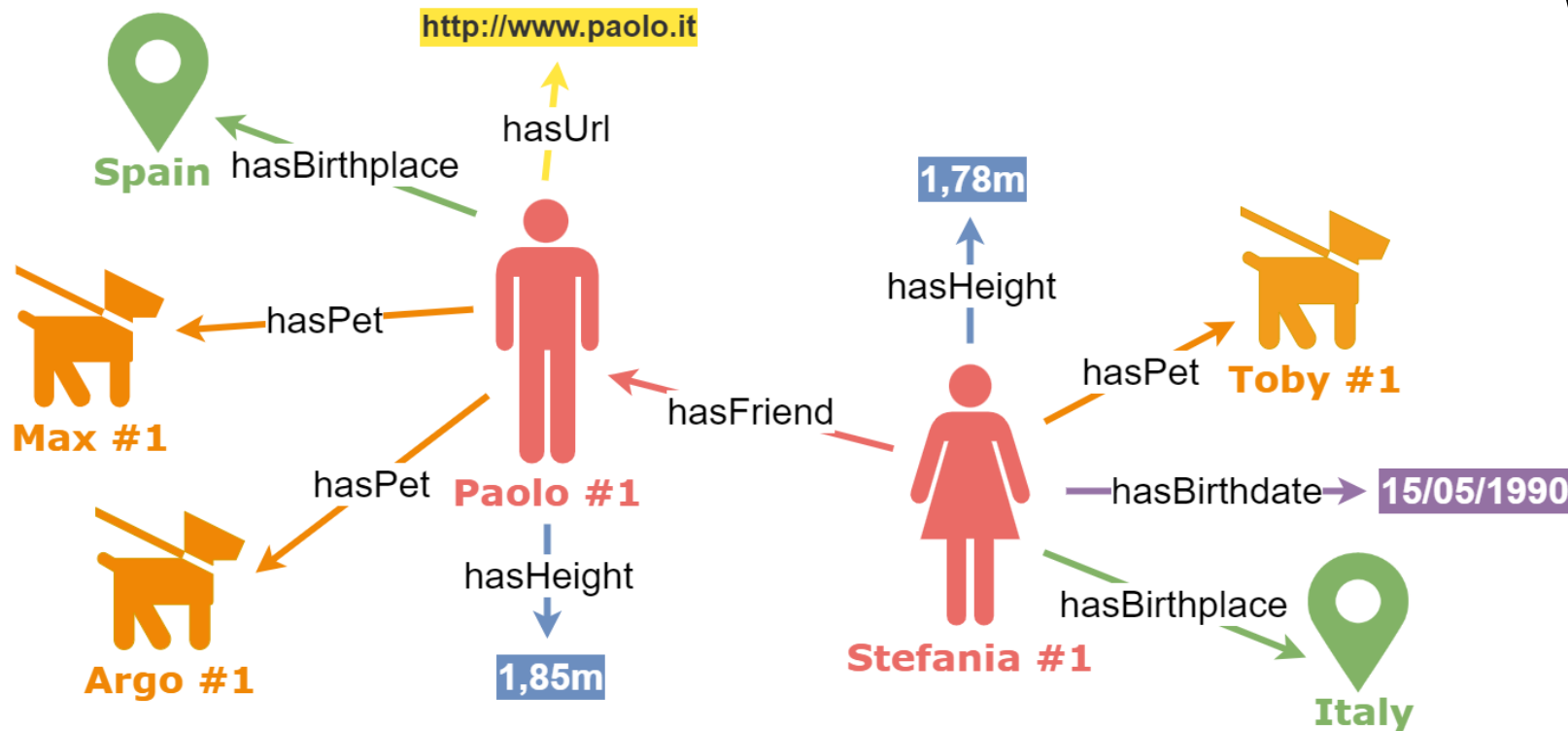
LoE – which (names of) percepts?

The **LoE world model** represents the following **Entity Graph (EG)** elements:

- An **Entity** is anything to which we give a name;
- An **Entity type (etype)** is the main class to which an entity belongs;
- A **Data Value (value)** is anything whose name and characteristics are predefined;
- A **Datatype** is a class of values;
- A **Data Property**, also called **Attributive Property** or also **Attribute***, describes an property type;
- An **Object Property** describes a relation type.

** Term derived from Relational DB terminology. It preserves the same intuition.*

An example of LoE EG



Which percepts?
Which facts?

Well-formedness conditions

An EG, to be well-formed must satisfy the following conditions:

- Each node is associated to one and only ONE entity or value;
- Each node is associated to one and ONLY one etype/ dtype;
- Each link is associated with one and only one data or object property;
- Data and object properties must have the correct etypes or datatypes (strong typing);
- No links are allowed starting from data value nodes.

LoE – The Logic of entities

- Intuition
- **Definition**
- Domain
- Language
- Interpretation function
- Theories
- Entailment
- Reasoning problems
- Key notions

LoE – The Logic of entities

Definition (LoE)

$$\text{LoE} = \langle \text{EG}, |=_{\text{LoE}} \rangle$$

with

$$\text{EG} = \langle L_{\text{LoE}}, D_{\text{LoE}}, I_{\text{LoE}} \rangle$$

When no confusion arises, we drop the subscripts.

LoE – The Logic of entities

- Intuition
- Definition
- **Domain**
- Language
- Interpretation function
- Theories
- Entailment
- Reasoning problems
- Key notions

Domain

Definition (Domain)

$$D = \langle U, \{C\}, \{R\} \rangle$$

where:

- $U = \{u\}$ is called the **universe (of interpretation)** of D .
- $\{u\}$ is a set of **units** $u_1, \dots, u_n$, for some n
- $\{C\}$ is a set of **classes** $C_1, \dots, C_m$ of units, for some m , with $C_i \subseteq U$
- $\{R\}$ is a set of **binary relations** $R_1, \dots, R_p$ between units, for some p , with $R_i \subseteq U \times U$

Observation (EG, Binary relations). To comply with the graph notation we restrict ourselves to binary relations.



Universe of interpretation

Definition (Universe of interpretation)

$$U = E \cup V$$

where:

- $U = \{u\}$ is the **universe of interpretation**;
- $E = \{e\} \subseteq \{u\}$ is the **entity universe**;
- $V = \{v\} \subseteq \{u\}$ is the **value universe**;
- $\{e\}$ and $\{v\}$ are disjoint.

Observation (Entity universe, value universe). E and V can be further divided into smaller sets which can be optionally defined to be mutually disjoint.

Universe of interpretation – entities and values

Observation (Entities). The notion of entity defined here captures the notion of entity informally introduced in the model theory. An entity is a thing that we perceive.

Example (Etype). Entity, person, artifact, vehicle, car, bus, woman, man, ..., professor, father, student, parent. Some are mutually disjoint, some are not, some are subsets of others.

Observation (Values of dtypes). Values are not real world entities. They do not exist in the world as we perceive it. They are **abstract entities**, so-called **mind constructs**, that are defined to capture certain characteristics of entities. They allow to **compare** certain characteristics of entities, thanks to their predefined definition of equality and properties. Some are mutually disjoint, some are not, some are subsets of others.

Example 1 (values of dtypes).

- Numbers allow to compare quantities (1, 2, 3, 4, ... are all entities, called numerals)
- Unities of measure (e.g., weight, height) when associated to numbers, allow to compare, specific aspects of entities
- Qualitative dtypes with qualitative values (hair color, beauty, empathy, clarity)
- Qualitative dtypes with a quantitative definition (e.g., color, mood)

Example 2 (Dtype). Real, long, Integer are subsets of number. They are all disjoint. Identifier is a subset of string. They are all subsets of the most general datatype, that is, data.

Classes

Definition (Class)

$$\{C\} = E_T \cup D_T$$

where:

- $E_T = \{E_i\}$ is a **set of etypes** E_i , with $E_i = \{e\} \subseteq E$
- $D_T = \{D_i\}$ is a **set of dtypes** D_i , with $D_i = \{v\} \subseteq V$

Observation (etype). For any etype E_i , $E_i \subseteq E$.

Observation (dtype). For any dtype D_i , $D_i \subseteq V$

Observation (etype, dtype). Not all subsets of E (of V) are etypes (dtypes). To become so, they must be identified as such, i.e., they must be given a name and defined to be so.

Binary relations

Definition (Binary relation)

$$\{R\} = O_R \cup A_R$$

where:

- $O_R = \{O_i\}$ is a **set of object properties** O_i , with $O_i \subseteq E_k \times E_j$
- $A_R = \{A_i\}$ is a **set of attributes** A_i , with $A_i \subseteq E_k \times D_j$

Observation (Object and data relations). Dtypes can only be leaves of an EG. Any such leaf value can be queried to obtain the desired information

Facts

Observation. LoE allows for the following facts:

- Every etype/ dtype and its argument is a fact
- Every relation R populated by its two arguments is a fact

Facts only have one of four possible forms:

- $e \in E$, that is, the entity e has etype E
- $v \in D$, that is, the value v has dtype D
- $\langle e_1, e_2 \rangle \in O_i$, that is, the object property O_i holds between the two entities e_1, e_2
- $\langle e, v \rangle \in A$, that is, the attribute A of entity e has value v

An example of domain of EG (continued)

We have the following example EG

$E = \{\#1, \#2, \#3, \#4, \#5, \#6, \#7, \#8, \#9, 1,85m, 1,78m, 15/05/1990, \text{http://www.paolo.it}\}$

$E_T = \{\text{Entity, Person}\}$

$D_T = \{\text{Data, Real, Boolean, String, Date, Length}\}$

$\{R\} = \{hF, hD, hH, hB, hL, hU\}$

from which we construct the following facts in the domain

$D = \{\#1 \in \text{Entity}, 1,85 \in \text{Real}, 1,85 \in \text{Data}, hF(\#1, \#1), \dots\}$

LoE – The Logic of entities

- Intuition
- Definition
- Domain
- **Language**
- Interpretation function
- Theories
- Entailment
- Reasoning problems
- Key notions

The language of assertions

Definition (Assertional language)

$$L_a = \langle A_a, FR_a \rangle = \{a\}$$

where:

- L_a is an **assertional language**
- A_a is an **alphabet** for **assertion formation**
- FR_a is a **set of formation rules**
- $\{a\}$ is the set of assertions obtained by the exhaustive application of FR_a to A_a (the transitive closure $FR_a(A_a)$ of FR_a applied to A_a).

Alphabet

Definition (Alphabet A_a)*

$$A_a = \langle E, \{T\}, \{P\} \rangle$$

where:

- $E = \{e\} \cup \{v\}$ is a set of (names of) **entities** e and of **values** v ;
- $\{T\} = \{E_i\} \cup \{D_i\}$ is a set of unary predicates standing for **etypes** and **dtypes**;
- $\{P\} = \{O_i\} \cup \{A_i\}$ is a set of binary **properties**, where O_i is an **object property**, also called a **role**, and A_i is an **attribute**.

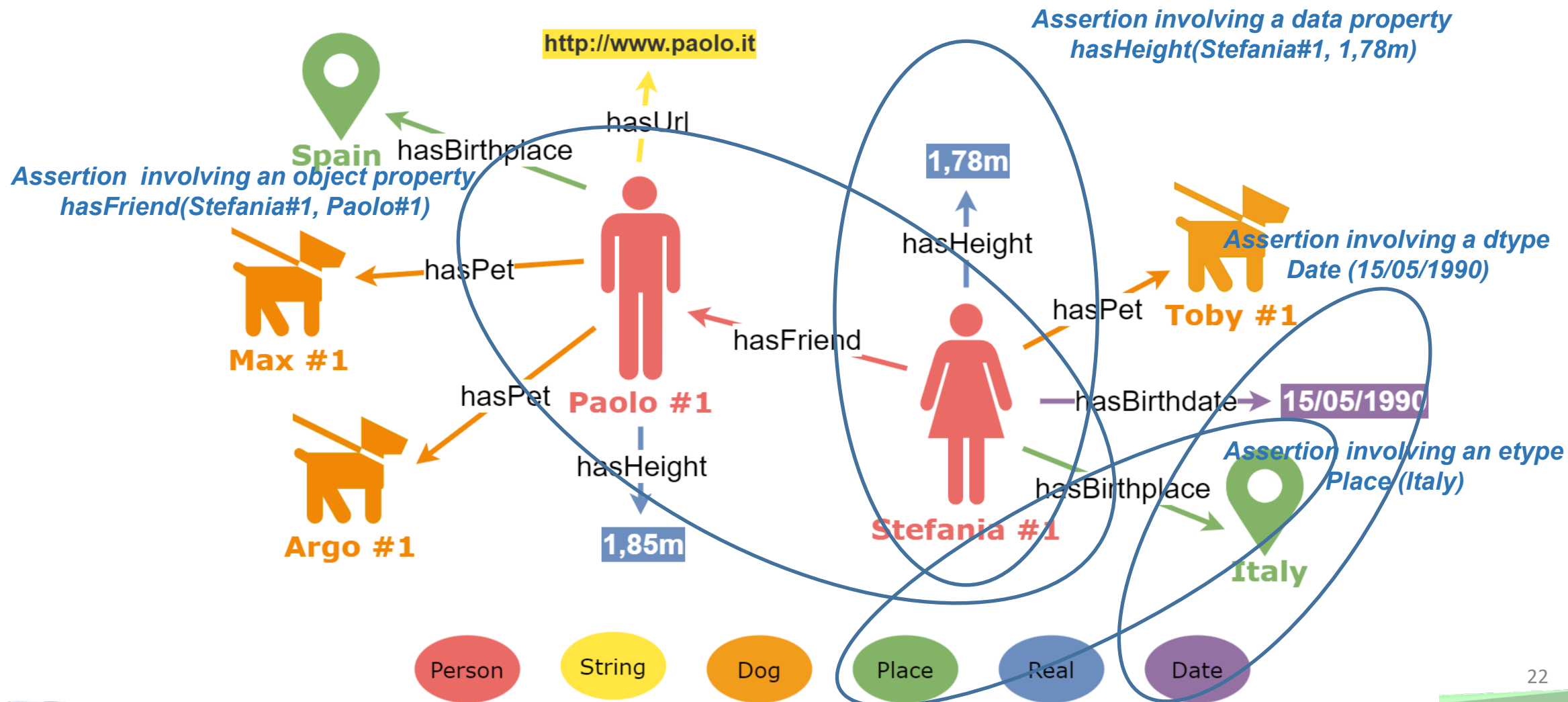
*The elements of the alphabet are written in *italic* to distinguish them from percepts

Formation Rules – BNF

$\langle \text{assertion} \rangle$	$::= \langle \text{etype} \rangle (\langle \text{nameEntity} \rangle$	
	$\langle \text{dtype} \rangle (\langle \text{value} \rangle$	
	$\langle \text{objProp} \rangle (\langle \text{nameEntity} \rangle, \langle \text{nameEntity} \rangle)$	
	$\langle \text{dataProp} \rangle (\langle \text{nameEntity} \rangle, \langle \text{value} \rangle)$	
$\langle \text{etype} \rangle$	$::= E_1 \mid \dots \mid E_n$	
$\langle \text{dtype} \rangle$	$::= D_1 \mid \dots \mid D_n$	
$\langle \text{objProp} \rangle$	$::= O_1 \mid \dots \mid O_n$	
$\langle \text{dataProp} \rangle$	$::= A_1 \mid \dots \mid A_n$	
$\langle \text{nameEntity} \rangle$	$::= e_1 \mid \dots \mid e_n$	
$\langle \text{value} \rangle$	$::= v_1 \mid \dots \mid v_n$	

Observation (BNF). This BNF generates assertions which are triples. It can be used to generate entity graph assertions as well as as a set of textual assertions.

An example of EG – assertions



Assertions – as from BNF (same as facts)

Observation. LOE allows for the following assertions:

- Every etype/ dtype and its argument is an assertion.
- Every relation R and its two arguments is an assertion.

Assertions only have one of four possible forms:

- $E_T(e)$, meaning that the entity e is of etype E_T
- $D_T(v)$, meaning that the value v is of dtype D_T
- $O(e_1, e_2)$, meaning that the object property O holds between e_1 and e_2
- $A(e, v)$, meaning that the data property A of entity e has value v

LoE expressiveness – observation

Observation 1 (LOE expressivity). The language of LoE is mapped one-to-one to the structure of domain interpretation.

Observation 2 (LOE expressivity). In LoE, given the low expressiveness of the language, the properties of etypes and dtypes can be exploited to make assertions but they cannot be reasoned about (similar to Relational DBs).

Example 1 (LoE weak expressivity: professor and person). The fact that Fausto is a person cannot be derived from the fact that he is a professor. There is no way to state the fact subsumption that all professors are persons (called a subsumption).

Example 2 (LoE weak expressivity: professor and person). One could think of solving the problem above by adding to each professor the fact that (s)he is also a person. This would still not allow to assert the general statement, instance-independent, that “a professor is also a person”.

LoE – The Logic of entities

- Intuition
- Definition
- Domain
- Language
- **Interpretation function**
- Theories
- Entailment
- Reasoning problems
- Key notions

Interpretation function

Definition (Interpretation function, alphabet). The LoE interpretation function I is defined as

$$I : L \rightarrow D$$

where I is defined as

$$I = \langle I_U, I_T, I_P \rangle$$

where:

- $I_U = \langle I_e, I_v \rangle$, the universe interpretation function, is the pair of the entity and the value interpretation function;
- $I_T = \langle I_E, I_D \rangle$, the Type interpretation function, is the pair of the etype and the dtypes interpretation function;
- $I_P = \langle I_O, I_A \rangle$, the Property interpretation function, is the pair of the object property and the attribute interpretation function

with:

$$I_e : \{e\} \rightarrow \{e\}$$

$$I_E : \{E_i\} \rightarrow \{E_i\}$$

$$I_O : \{O_i\} \rightarrow \{O_i\}$$

$$I_v : \{v\} \rightarrow \{v\}$$

$$I_D : \{D_i\} \rightarrow \{D_i\}$$

$$I_A : \{A_i\} \rightarrow \{A_i\}$$

Interpretation function (continued)

Definition (LoE interpretation function, assertion). The interpretation of assertions into facts is defined as follows:

$$\begin{aligned} I(E_i(e)) &= I_E(E_i)(I_e(e)) &= e \in E_i \\ I(D_i(v)) &= I_D(D_i)(I_v(v)) &= v \in D_i \\ I(O_i(e_1, e_2)) &= I_O(O_i)(I_e(e_1), I_e(e_2)) &= \langle e_1, e_2 \rangle \in O_i \\ I(A_i(e, v)) &= I_A(A_i)(I_e(e), I_v(v)) &= \langle e, v \rangle \in A_i \end{aligned}$$

Observation. Notice how the interpretation of assertions first applies the interpretation function (i.e., I) which then, **compositionally**, applies the proper component specific interpretation function mimicking, step by step how syntax composes (e.g., in the first assertion, I_E and I_e).

Observation. The definition $I = \langle I_E, I_T, I_P \rangle$ is the **intensional** view of the interpretation function. The definition of how it applies to assertions (this page) is its **extensional** view.

Notation. The words of the alphabet are in “*italic*”, percepts are in “roman”

Interpretation function – Example (continued)

We have the following

$I(\text{Paolo})$	$= \#1$
$I(\text{hasFriend}(\text{Paolo}, \text{Sofia}))$	$= \text{hf}(\#2, \#1)$
$I(\text{Person}(\text{Stefania}))$	$= P(\#3)$
$I(\text{hasDog}(\text{Stefania}, \text{Toby}))$	$= \text{hD}(\#3, \#6)$
$I(\text{hasDog}(\text{Paolo}, \text{Argo}))$	$= \text{hD}(\#1, \#4)$
$I(\text{hasDog}(\text{Paolo}, \text{Max}))$	$= \text{hD}(\#1, \#5)$

Interpretation function - compositionality

Observation (Semantic compositionality). By semantic compositionality we mean the fact that the interpretation function interprets the elements of the alphabet one by one and builds the resulting final fact by composing the resulting percepts by mimicking how the formation rules compose.

In this way, either an input assertion is not well-formed, in which case it makes no sense to ask what it means, or the resulting fact will be the right interpretation of the input assertion, by construction.

This is how interpretation functions are built, for all logical languages sentences, not only for assertional languages and assertions.

It guarantees correctness, and it is also easily understandable by humans, as this is also how the semantics of natural languages are built.

LoE – The Logic of entities

- Intuition
- Definition
- Domain
- Language
- Interpretation function
- **Theories**
- Entailment
- Reasoning problems
- Key notions

LoE theories

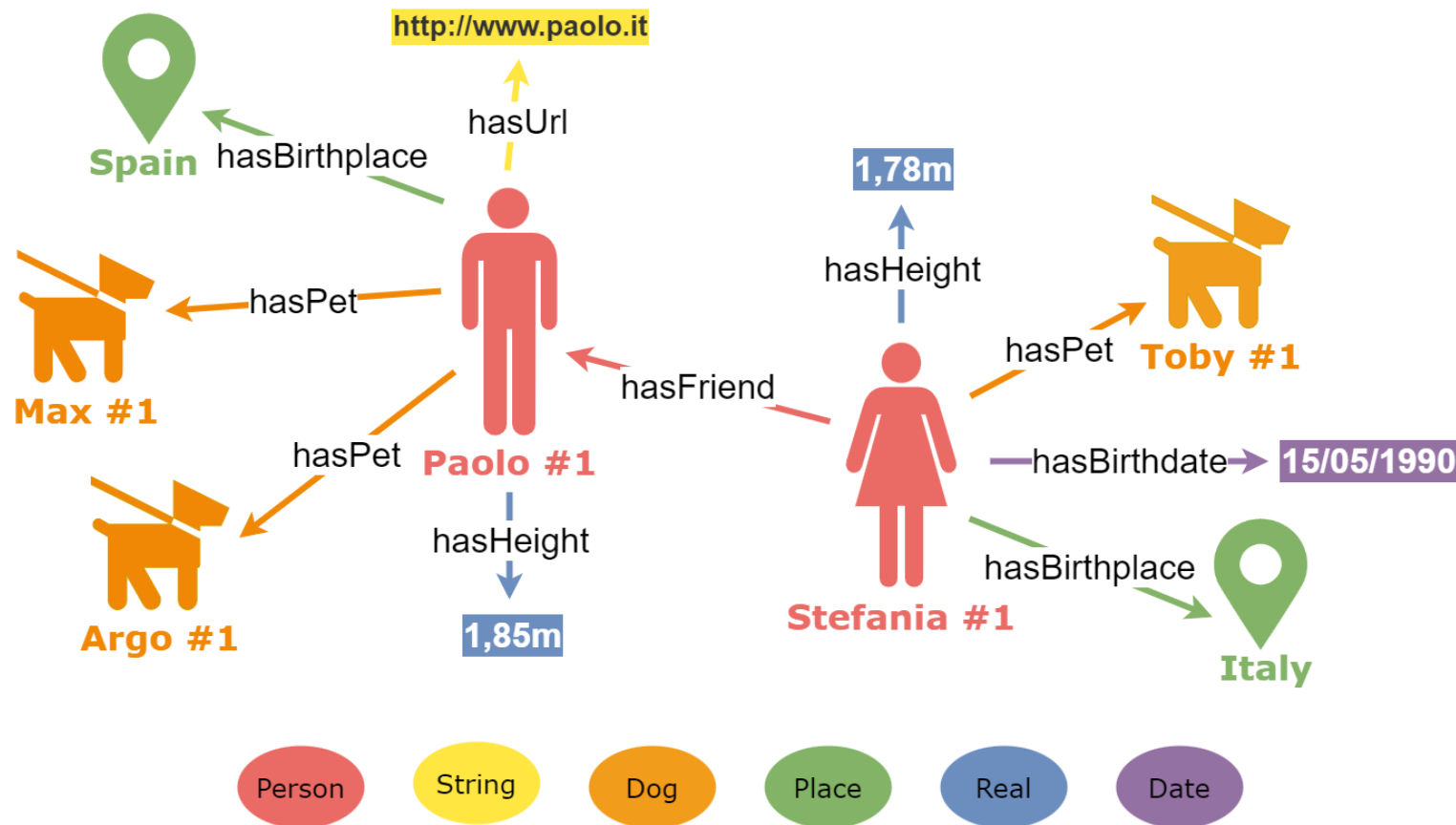
Definition (LoE theory). A **LoE theory** $T_{\text{LoE}} = \{a\} \subseteq L_{\text{LoE}}$ is a set of LoE assertions, where a LoE assertion a has one of four possible forms:

- $E_T(e)$, meaning that the entity e is of etype E_T
- $D_T(v)$, meaning that the value v is of dtype D_T
- $O(e_1, e_2)$, meaning that the object property O holds between e_1 and e_2
- $A(e, v)$, meaning that the data property A of entity e has value v

Proposition (LoE theory). Any LoE theory is canonical.

Proposition evidence (LoE Theory). By construction for any assertion in a LoE theory there is one and only fact in the model which is the result of the application of the interpretation function to the input assertion.

LoE Theory – exercise 1



Exercise. Build the LoE theory of the EG above

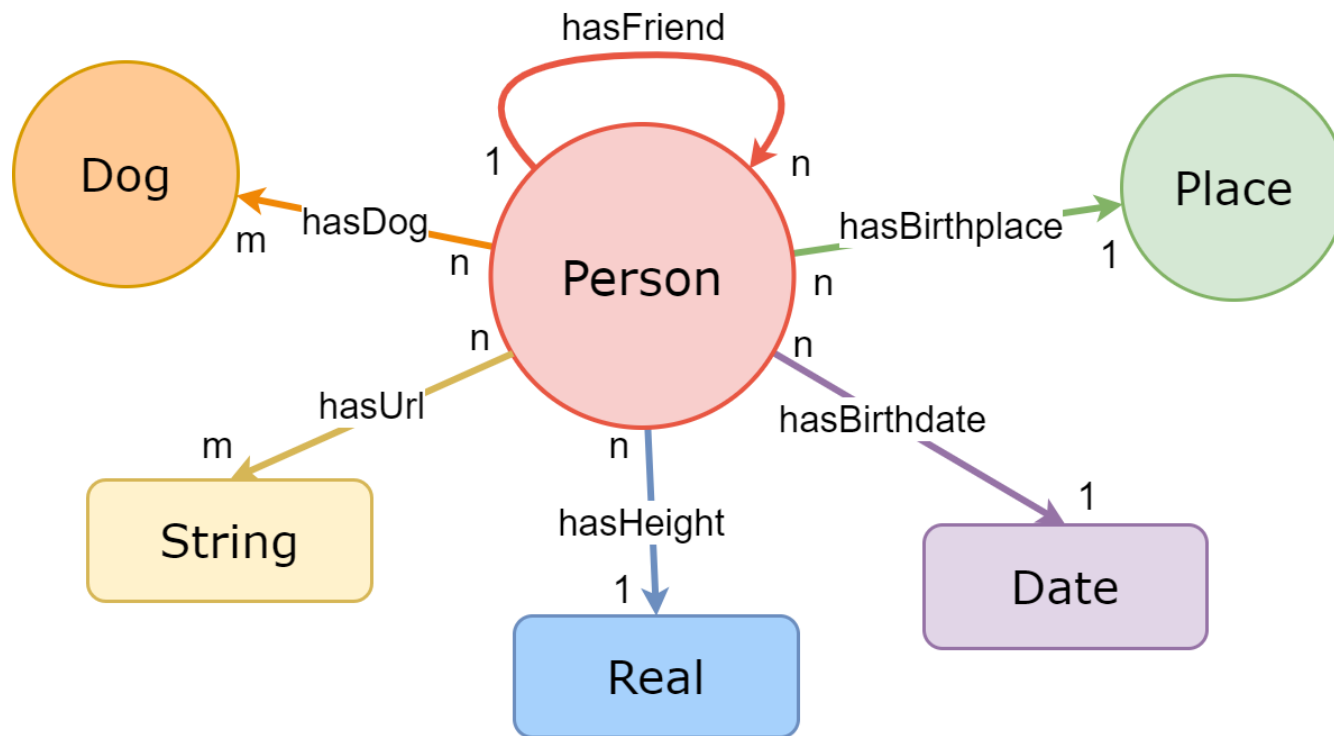
LoE Theory – exercise 1

Alphabet

- **Set of entities** = {Paolo, Sofia, Stefania, Argo, Max, Toby, Balto, Spain, Italy, Balto Spain, Italy}
- **Set of values** = {1,85m, 1,78m, 15/05/1990, <http://www.paolo.it>}
- **Set of etypes** = {Person, Dog, Place}
- **Set of dtypes** = {Measurement, Date}
- **Set of object properties** = {hasFriend, hasDog, hasBirthPlace}
- **Set of data properties** = {hasBirthdate, hasUrl, hasHeight}

Assertions = {Person(Paolo), hasBirthplace(Paolo, Spain)
hasHeight(Paolo, 1,85m), hasDog(Paolo, Argo),
hasUrl(Paolo, <http://www.paolo.it>), hasDog(Paolo, Max)}

LoE Theory – exercise 2



Exercise. Build the LoE theory of the TLO above assuming that the etypes have only one instance with an unknown name.

Language, Theory - Correctness, completeness

Proposition (Language correctness and completeness). L_{LOE} is correct with respect to D_{LOE} . L_{LOE} is complete with respect to D_{LOE} if I_{LOE} is surjective.

Proposition evidence (Language correctness and completeness). From the definition of L_{LOE} and I_{LOE} for any assertion there is a corresponding fact in the domain, not necessarily only one assertion per fact. We have completeness when the alphabet contains symbols for all the percepts in the domain. Which is guaranteed with canonical representations.

Proposition (Theory correctness and completeness). A theory $T_a = \{a\}$ such that, if $a \in T_a$ then $I(a) \in M$ is correct with respect to M . T_a is complete only if it is maximal.

Proposition evidence (Theory correctness and completeness). By construction. For any correct assertion, the compositionality of the interpretation function guarantees that the correct percept is computed. Completeness is guaranteed with canonical representations.

LoE – The Logic of entities

- Intuition
- Definition
- Domain
- Language
- Interpretation function
- Theories
- **Entailment**
- Reasoning problems
- Key notions

World entailment (Recall)

Definition (World entailment, satisfiability) Let $W = \langle L_a, D, I_a \rangle$ be a world model. Let L_a be an assertional language. Then $\models_{L_a}$ is a **world entailment relation**

$$\models_{L_a} \subseteq D \times L_a$$

such that

$$M \models_{L_a} T_a$$

$M = \{f\} \subseteq D$ is a **model of** $T_a \subseteq L_a$. We say that M **entails** T_a . We write $M \models T_a$ instead of $M \models_{L_a} T_a$ when no confusion arises.

Logical entailment (Recall)

Definition (Logical entailment). Let $L_W = \langle W, \models_W \rangle$ be a world logic with $W = \langle L, D, I \rangle$ its world model. Let $\{M\} \subseteq D$ be a set of models of T with $T, \underline{I} \subseteq L$ and $a \in L$. Then we write

$$T \models_{\{M\}} \underline{I} \qquad (T \models_{\{M\}} a)$$

and say that T **logically entails** $\underline{I}$ (T **logically entails** a) with respect to $\{M\}$ if

$$\text{for all } M \in \{M\}, \text{ if } M \models T \text{ then } M \models \underline{I} \text{ (} M \models a \text{)}$$

Observation (Logical entailment, reasoning). Logical entailment implements reasoning in the following way: “Whatever we derive from T must be true in the same sets of “relevant” models $\{M\}$ as T .”

Observation (Which set of models $\{M\}$). The choice of the models $\{M\}$ of T is arbitrary. It could be the only model of T , all models of T , a model selected because it is the intended model of T , and so on. If $M \not\models T$ then Logical entailment does not apply. T has the key role of selecting which models $M \in \{M\}$ must be considered.

Notation ($\{M\}$). When no confusion arises we write $\models$ instead of $\models_{\{M\}}$.

LoE Entailment

Proposition (LoE entailment). Let T_a be a LoE theory for a model M , with $T_a \subseteq L_a$. Let $\underline{a} \in L_a$ and $T_{\underline{a}} \subseteq L_a$ be respectively an **assertion question** and a **theory question**. Then:

$M \models T_{\underline{a}}$ if and only if $T_a \models T_{\underline{a}}$ if, for all $a \in T_{\underline{a}}$, $a \in T_a$

$M \models \{\underline{a}\}$ if and only if $T_a \models \{\underline{a}\}$, written $T_a \models \underline{a}$, if $\underline{a} \in T_a$

Proposition evidence (LoE entailment). The LoE world model is canonical. There is a one-to-one mapping between a LOE theory and a LoE model.

Observation (LoE entailment). The LoE world entailment can be seen as translating two questions: (1) the default one, whether $\underline{a}$ is true in the reference (world) model and (2), whether $\underline{a}$ was already known and as such, it was already in memory.

Observation (Truth, Falsity and memory). Based on the previous observation, saying that something is True amounts to saying that your memory says so. No real reasoning involved.

Entailment – Decidability, complexity

Proposition (Entailment correctness, completeness). LoE entailment is complete with respect to the canonical model.

Observation (Entailment correctness, completeness). LoE entailment is incomplete with respect to: (i) the model which considers also the meaning encoded in the reference KTLO generating the schema, and with respect to (ii) the model which considers the CG encoding the meaning of the concepts used in a LoE theory.

Proposition (LoE decidability, complexity). LoE entailment is decidable in polynomial time.

Proposition evidence (LoE decidability, complexity). Entailment can be computed by sequentially checking the equality of the input formula (the candidate theorem) with all the axioms of the LoT theory.

Observation (Weakness of LoE entailment). The language of LoE, like that of any informal or semi-formal world model, has very low expressibility, and it allows to make ONLY **positive** and **partial assertions** about what is the case. For instance, it does not provide any means for saying what is NOT the case. Or that two assertions are mutually inconsistent.

Example (Weakness of LoE entailment). Thus, for instance, in a EG one can add the two assertions ON(pen, table) and ON(table, pen). These two assertions are intuitively inconsistent but the language of LoE does not have the expressibility for asserting this fact. Any LoE theory can be extended with all the assertions allowed by the language.

LoE – The Logic of entities

- Intuition
- Definition
- Domain
- Language
- Interpretation function
- Theories
- Entailment
- **Reasoning problems**
- Key notions

LoE Instance checking

Definition (LoE Instance checking) LoE instance checking checks whether an assertion is entailed by a LoE theory. We have the following four possible cases

$T \models E(e)$	returns yes	if $E(e) \in T_{\text{LoE}}$,	IDK otherwise
$T \models D(v)$	returns yes	if $D(v) \in T_{\text{LoE}}$,	IDK otherwise
$T \models O(e_1, e_2)$	returns yes	if $O(e_1, e_2) \in T_{\text{LoE}}$,	IDK otherwise
$T \models A(e, v)$	returns yes	if $A(e, v) \in T_{\text{LoE}}$,	IDK otherwise

Observation 1 (LoE Instance checking). It amounts to checking whether the input assertion, modulo synonyms, is in the theory which generated the model.

LoE Instance checking – closed world

Observation (Instance checking). Consider the following two variations of instance checking

$T \models E(e)$ returns “yes, $E(e)$ is true” if $E(e) \in T_{LoE}$
returns “IDK” otherwise

$T \models E(e)$ returns “yes, $E(e)$ is true” if $E(e) \in T_{LoE}$
returns “no, $E(e)$ is false” otherwise

Open World Assumption (OWA): The first version, LoE instance checking, returns «I don’t know» when entailment does not hold.

Closed World Assumption /CWA): The second version, returns that « $E(e)$ is false» when entailment does not hold.

Observation. CWA transforms **partial knowledge** (a fact not being in the model) into **negative knowledge**. What is not known is assumed to be false. OWA is mainly applied in the Web, CWA is mainly applied corporate DBs.

LoE – Instance retrieval

Definition (LoE Instance retrieval). Given an etype (or object/ data property), LoE Instance retrieval returns all the entities (or pair “entity, entity/data”) in the memory T which satisfy the etype (or object property/attribute) in input

$$T \models E$$

$$T \models D$$

$$T \models O$$

$$T \models A$$

Observation (Instance retrieval). It generates all the possible assertions using the entities and values of the alphabet and applies model checking to the resulting assertion. It returns the assertions for which model checking returns “yes”.

Observation (Instance retrieval). Instance retrieval consists of a multiple call to Instance checking. Similar to Q/A in relational DBs.

LoE – The Logic of entities

- Intuition
- Definition
- Domain
- Language
- Interpretation function
- Theories
- Entailment
- Reasoning problems
- **Key notions**

Key notions

- LoE as a logic of entities
- Entity graphs (EGs)
- Entities and values
- Etypes and dtypes
- Object properties
- Data properties (alias attributes)
- LoE alphabet
- LoE formation rules
- LoE assertions
- LoE expressiveness
- Interpretation function
- Intensional definition of an interpretation function
- Extensional definition of an interpretation function
- Entailment relation
- LoE Instance checking
- OWA
- CWA
- LoE Instance retrieval



UNIVERSITY
OF TRENTO - Italy

Dipartimento di Ingegneria e Scienza dell'Informazione



LoE

The Logic of Entities

(HP2T)

Oct 13, 2023